Modified

CBCS SCHEME

USN | | | | | | | | | | |

21CS44

Fourth Semester B.E. Degree Examination, June/July 2023
Operating System

Time: 3 hrs. Max. Marks: 100

Note: ***Answer any FIVE full questions, choosing ONE full question from each module.***

Module-1

1 a. With neat diagram, explain :
i) Operating System ii) Dual Mode Operation in O.S. **(08 Marks)**

b. Explain the Operating System services with respect to programs and the users. **(06 Marks)**

c. What is a Process? Explain different states of a process with state diagram. **(06 Marks)**

OR

2 a. With neat diagram, explain the concept of Non virtual machine, Virtual machine and VMware architecture. **(10 Marks)**

b. Define : i) Context switching ii) Direct and Indirect communication
iii) Automatic and Explicit buffering. **(10 Marks)**

Module-2

3 a. What is Multithreaded process? Explain four benefits of Multithreaded programming. **(06 Marks)**

b. Consider the set of process with Arrival time, CPU burst time (in milliseconds) and priority shown below (Lower number represents higher priority).

Process	Arrival Time	Burst Time	Priority
P1	0	10	3
P2	1	1	1
P3	2	2	4
P4	3	1	5
P5	4	5	2

Write the Gantt chart and solve the Average waiting time and Average turnaround time for
i) SJF Scheduling (Preemptive) ii) Priority Scheduling (Preemptive).
(NOTE : Consider Arrival Time for both Algorithms). **(14 Marks)**

OR

4 a. Explain with diagram : i) Multithreading models ii) Multilevel queue scheduling. **(08 Marks)**

b. What is Critical – Section? How do you implement a monitor solution to the dining – philosophers problem? **(12 Marks)**

Module-3

5 a. What is a Deadlock? What are the four necessary conditions for the deadlock to occur? **(04 Marks)**

b. What are the two methods to eliminate deadlock? **(02 Marks)**

c. Consider the following snapshot of a system :

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P1	2	0	0	1	4	2	1	2	3	3	2	1
P2	3	1	2	1	5	2	5	2				
P3	2	1	0	3	2	3	1	6				
P4	1	3	1	2	1	4	2	4				
P5	1	4	3	2	3	6	6	5				

Answer the following using Banker's algorithm.
i) Is the system is in safe state? If so, what is the safe sequence?
ii) If request from process P2 (0, 1, 1, 1) is considered immediately, what is the System state and Sequence? **(14 Marks)**

OR

6 a. Which are the commonly used strategies to select a free hole from the available holes? **(06 Marks)**
b. With suitable diagram, explain external fragmentation. **(04 Marks)**
c. With neat diagram, explain paging hardware with TLB. **(10 Marks)**

Module-4

7 a. What is Demand Paging? Explain the steps in handling page fault using appropriate diagram. **(10 Marks)**
b. Consider the page reference string : 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 for a memory with 3 frames. Determine the number of page faults using Optimal and LRU replacement algorithms. Which algorithm is most efficient? **(10 Marks)**

OR

8 a. With neat diagram, explain Two – level and Three – level directory structure. **(08 Marks)**
b. Explain Contiguous and Linked disk space allocation methods with diagram. **(12 Marks)**

Module-5

9 a. A drive has 200 cylinders 0 to 199. Head starts at 53 to serve the request queue : 98, 183, 37, 122, 14, 124, 65, 67. Draw disk head schedule diagram and explain for FCFS , SSTF, C – SCAN and C – LOCK. **(12 Marks)**
b. How the Access matrix model of protection can be viewed in OS? **(08 Marks)**

OR

10 a. With neat diagram, explain SAN and MULTICS. **(08 Marks)**
b. Explain the components of a Linux System. **(06 Marks)**
c. Explain in brief fork ( ) and exec ( ) system calls in Linux / UNIX OS, also write a program to implement these system calls in C language. **(06 Marks)**

* * * * *

Re: Sir/Madam, regarding scheme modifications

"Nagappa Bhajantri" <bhajan3nu@gmail.com>

October 13, 2023 4:53 PM

To: boe@vtu.ac.in

CC: bhajan3@gmail.com

Sir,
Here mentioned course 21CS44 Operating System is not required any modification in scheme, scheme is correct & updated

Thanking you

On Fri, Oct 13, 2023 at 2:00 PM <boe@vtu.ac.in> wrote:

--
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Professor & HOD, Computer Science & Engg Dept.,
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Karnataka-571313, cell:94486 74093

" APPROVED "

Registrar (Evaluation)
Visvesvaraya Technological University
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Visvesvaraya Technological University

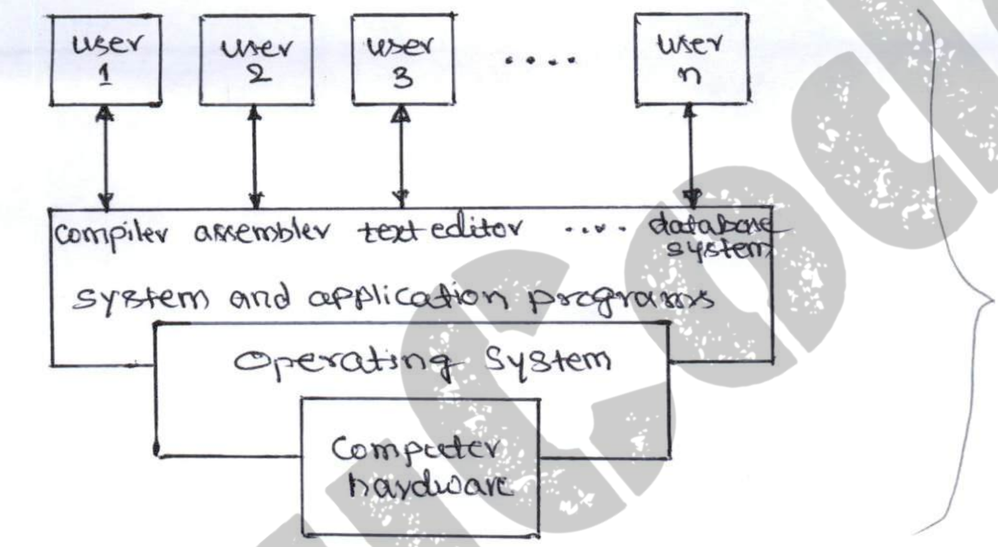
Belagavi, Karnataka - 590 018.

Scheme & Solutions

X Signature of Scrutinizer

Subject Title : Operating System **Subject Code :** 21CS44

Question Number	Solution	Marks Allocated
1 a)	(i) Operating System (ii)	8 (2+2+2+2)



(2) M

*) An operating system is acts as an intermediary between the computer user and the computer hardware.

*) hardware : central processing unit (CPU), memory, and input/output devices

*) application programs : word processors, spreadsheets, compilers, web browsers etc.

(2) M

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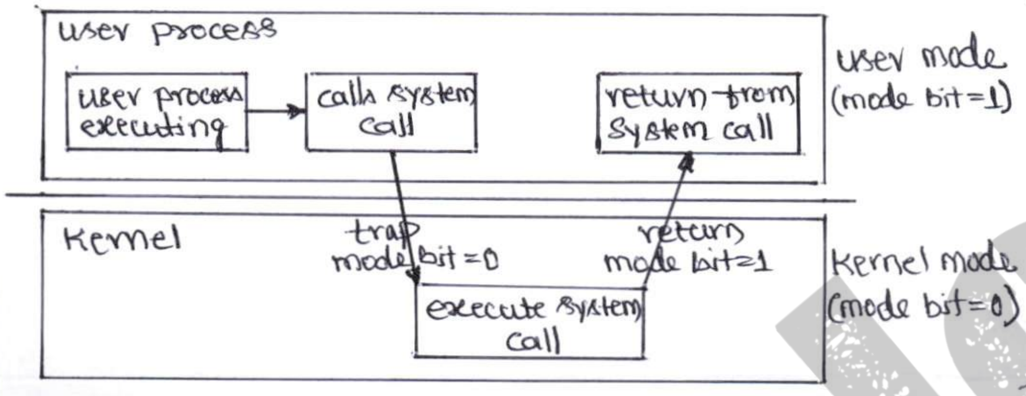
Registrar (Evaluation)
Visvesvaraya Technological University
BELAGAVI - 590018

Question Number	Solution	Marks Allocated

(ii) Dual mode operation in O.S



②M

*) hardware of the computer to indicate the current mode } mode bit < Kernel(0) user(1)

*) user mode : the computer system is executing on behalf of a user application.

*) Kernel mode : the computer system is executing on behalf of operating system.

②M

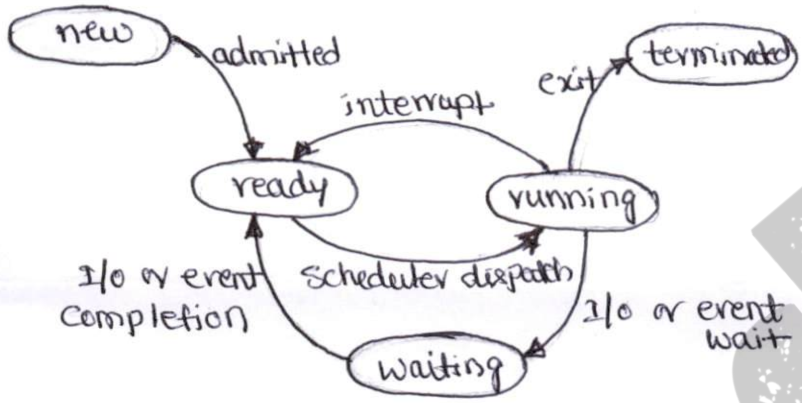
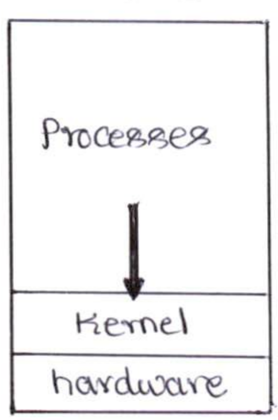
1 b)

6 (1+1+1+1+1+1)

*) User interface
*) Program execution
*) I/O operation
*) File system manipulation
*) Communications
*) Error detection
*) Resource allocation
*) Accounting
*) Protection & security

any six, one mark each with explination.

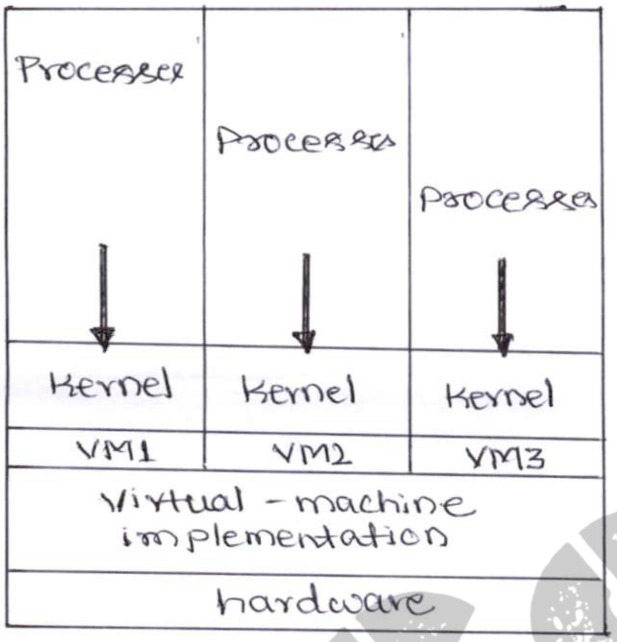
Subject Title : Operating System   Subject Code : 21CS44

Question Number	Solution	Marks Allocated
1 c)	Process is a program in execution. } 1 M	6 (1+3+2)
	 	3 M
	*> New: The process is created *> Running: Instructions are being executed. *> Waiting: Process is waiting for some event to occur (I/O completion or signal) *> Ready: Process is waiting to be assigned to a processor. *> Terminated: The process finished execution.	2 M
2 a)	Nonvirtual machine	10 (2+4+4)
	 with explination	2 M

Subject Title : Operating System Subject Code : 21CS44

Question Number	Solution	Marks Allocated

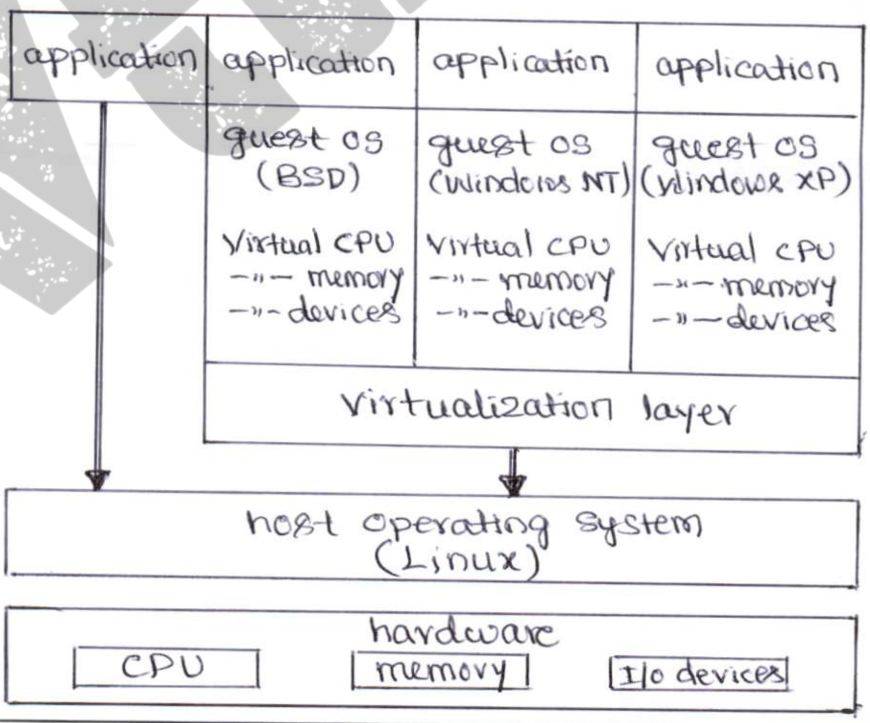
Virtual machine



*) The fundamental idea behind a virtual machine is to abstract the hardware of a single computer into several different execution environment.

④ M

VMware architecture

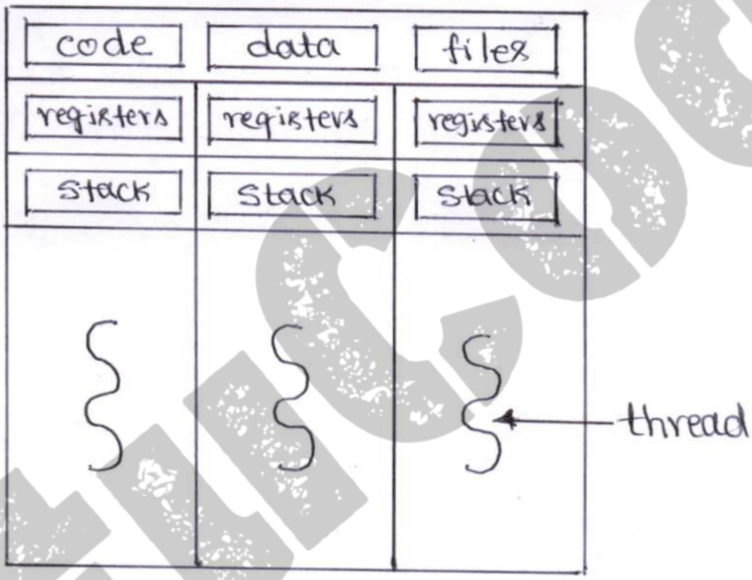


④ M

PTO

Subject Title : Operating System Subject Code : 21CS44

Question Number	Solution	Marks Allocated
	*> VMware runs as an application on a host OS such as Windows or Linux and allows this host system to concurrently run several different guest OS's as independent virtual machines. *> Explination of the diagram.	
2 b)	(i) Context switching (ii) (iii)	10 (2+4+4)
	*> Switching the CPU to another process requires state save the current process and state restore a different process. This task is known as context switch.	2 M
	(ii) Direct and indirect communication *> In "direct communication" process must explicitly name the recipient or sender of the communication. send (P, message) – Send message to process P. receive(Q, message) – Receive message from process Q. *> In "indirect communication" messages are sent to and receive from "mailboxes" or "ports". send (A, message) – send message to mailbox A. receive (A, message) – receive message from mailbox A.	4 M

Question Number	Solution	Marks Allocated
	(iii) Automatic and explicit buffering *) Messages exchanged by communicating processes reside in a temporary queue (buffering) *) Zero capacity buffering *) Bounded —— " —— *) unbounded —— " —— explination of each	4 M
3 a)	 *) If a process has multiple threads of control, it can perform more than one task at a time.	6 (3+3) 3 M
	*) Benefits (i) Responsiveness (ii) Resource sharing (iii) Economy (iv) Utilization of multi-processor architectures. explination of each	3 M

Subject Title : Operating System   Subject Code : 21CS44

Question Number	Solution	Marks Allocated

3 b) **(i) SJF : Gantt chart** — **14 (7+7)**

P1	P2	P3	P4	P5	P1

0 1 2 4 5 10 19 — 3 M

Waiting Time (WT)

P1 = 10 − 1 = 9
P2 = 1 − 1 = 0
P3 = 2 − 2 = 0
P4 = 4 − 3 = 1
P5 = 5 − 4 = 1

Avg W.T = $\frac{11}{5}$ = 2.2 ms — 2 M

7 M (3+2+2)

Turnaround Time (T.T = WT + BT)

P1 = 9 + 10 = 19
P2 = 0 + 1 = 1
P3 = 0 + 2 = 2
P4 = 1 + 1 = 2
P5 = 1 + 5 = 6

Avg T.T = $\frac{30}{5}$ = 6 ms — 2 M

(ii) Priority : Gantt chart

P1	P2	P1	P5	P1	P3	P4

0 1 2 4 9 16 18 19 — 3 M

Waiting Time (WT)

P1 = 1 + 5 − 0 = 6
P2 = 1 − 1 = 0
P3 = 16 − 2 = 14
P4 = 18 − 3 = 15
P5 = 4 − 4 = 0

Avg WT = $\frac{35}{5}$ = 7 ms — 2 M

7 M (3+2+2)

Turnaround Time (TT = WT + BT)

P1 = 6 + 10 = 16
P2 = 0 + 1 = 1
P3 = 14 + 2 = 16
P4 = 15 + 1 = 16
P5 = 0 + 5 = 5

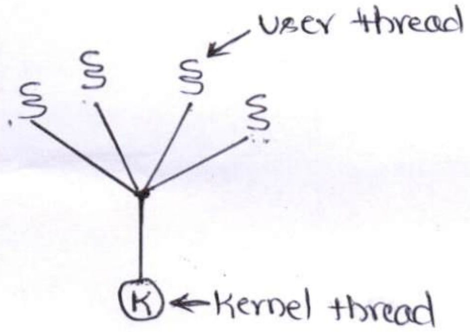
Avg TT = $\frac{54}{5}$ = 10.8 ms — 2 M

Subject Title : Operating System　　　　Subject Code : 21CS44

Question Number	Solution	Marks Allocated
4 a)	(i) (ii) Multithreading models	8 (4+4)

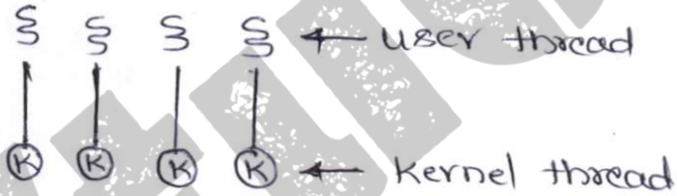
Following relationship exists between user threads and Kernel threads.

*> Many - to - One model :



Only one thread can access the Kernel at a time, multiple threads are unable to run in parallel on multi-processors.

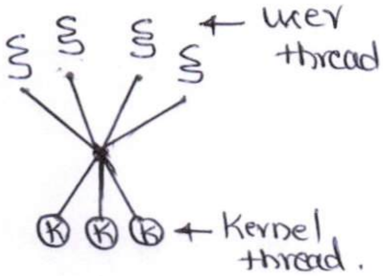
*> One - to - One model :



Maps each user thread to a Kernel thread. It provides more concurrency also allows multiple threads to run in parallel on multiprocessor.

④ M

*> Many - to - Many model :

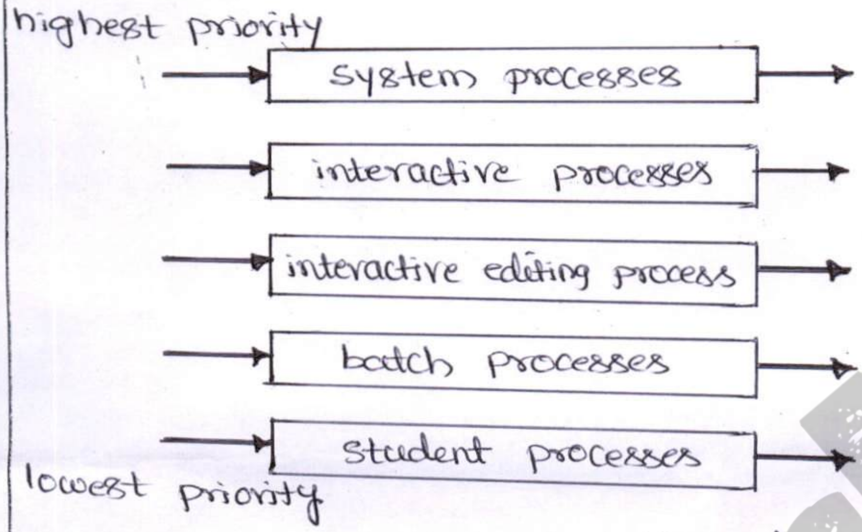


User can create as many user threads to a smaller or equal number of Kernel threads. Also the Kernel can schedule another thread for execution.

Subject Title : Operating System   Subject Code : 21CS44

Question Number	Solution	Marks Allocated

(ii) Multilevel queue Scheduling



A multilevel queue scheduling algorithm Partitions the ready queue into several seperate queue based on some priority. Low priority processes are executed when no processes are in higher priority queue. — ④ M

4 b) — 12 (2+10)

A system consisting of n processes $\{P_0, P_1, \ldots P_{n-1}\}$. Each process has a segment of important code called a "critical section".

```
do {
    [entry section]
        critical section
    [exit section]
        remainder section
} while (TRUE);
```

② M

Subject Title : Operating System **Subject Code :** 21CS44

Question Number	Solution	Marks Allocated
	*) Implementing monitor to dining-Philosopher problem.	10 M

```
monitor dp
{
  enum {THINKING, HUNGRY, EATING} state[5];
  condition self [5];
  void pickup (int i) {
      state [i] = HUNGRY;
      test [i];
      if state[i] != EATING)
          self [i]. wait();
  }
  void putdown (int i) {
      State [i] = THINKING;
      test ((i+4) % 5);
      test ((i+1) % 5);
  }
  void test (int i) {
      if ((State[(i+4) %5] != EATING) &&
         (State [i] == HUNGRY &&
         (State [(i+1) %5 != EATING) {
             State [i] = EATING;
             Self [i]. signal();
         }
  }
  initialization. code () {
      for (int i=0; i<5; i++)
          State [i] = THINKING;
  }
}
```

(*) **NOTE :-** Priority is given to the above code. (importance)

Subject Title : Operating System Subject Code : 21CS44

Question Number	Solution	Marks Allocated
5 a)	*) A process requests resources; if not available, process enters waiting state, waiting process never change state because the resources it has requested are held by other waiting processes. This situation is called a "deadlock". *) Necessary conditions (i) Mutual exclusion (ii) Hold and wait (iii) No preemption (iv) Circular wait. explination of these	4 4 M
5 b)	*) Two methods to eliminate deadlocks (i) Process termination (ii) Resource preemption Explination of these	2 M
5 c)	*) Need A B C D P1 → 2 2 1 1 P2 → 2 1 3 1 P3 → 0 2 1 3 P4 → 0 1 1 2 P5 → 2 2 3 3 4 M *) Yes, the system is in safe state. *) Safe sequence < P1, P4, P5, P2, P3 > 3 M	14 (7+7) 7 M

Subject Title : Operating System **Subject Code :** 21CS44

Question Number	Solution	Marks Allocated
	(ii) Request → P2 (0, 1, 1, 1)	

	Allo (A B C D)	Max	Availa	Need
P1 →	:	:	3 2 1 0	:
P2 →	3 2 3 2	5 2 5 2		2 0 2 0
:	:	:		:

*) If the process P2's additional request (0,1,1,1) is granted immediately, then the system went to "unsafe state", so "no safe sequence". — 7 M

6 a) — 6

(i) First fit
(ii) Best fit
(iii) Worst fit

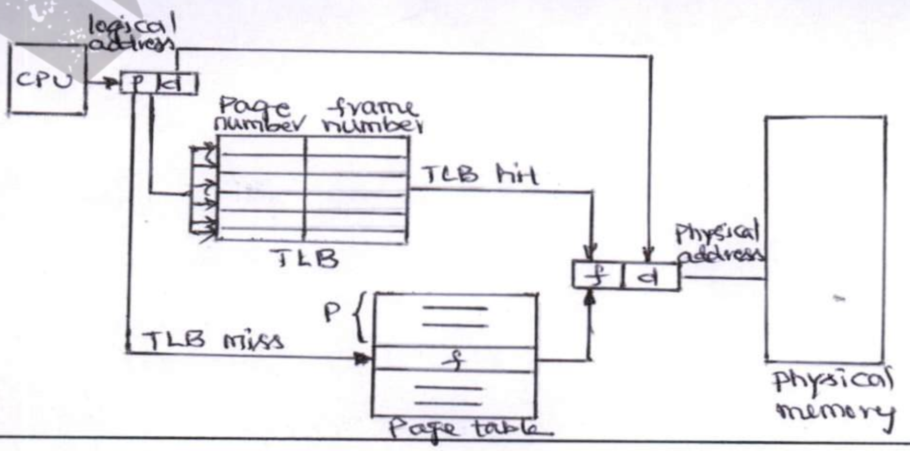
with explination 2 M × 3 = 6 M

6 b) — 4

Any suitable diagram } 2 M

Explination } 2 M

6 c) — 10 (5+5)



5 M

*) Explination — 5 M

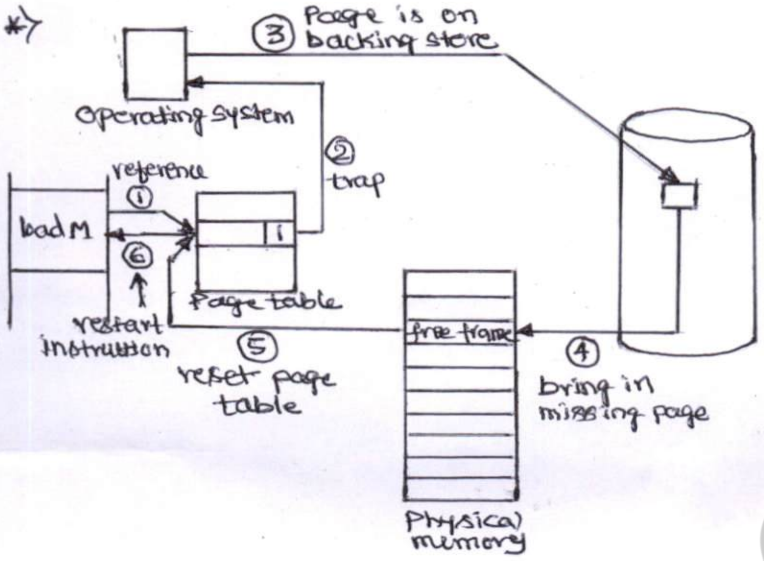
Subject Title : Operating System Subject Code : 21CS44

Question Number	Solution	Marks Allocated
7 a)	*) Load the pages only as they are needed. } ① M	10 (1+6+3)
	*) (diagram below) — ⑥ M	
	*) Above 6 steps explination. — ③ M	
7 b)	(i) Optimal — ⑤ M ← Efficient	10 (5+5)
	→ 9 page faults — ⑤ M	
	(ii) LRU — ⑤ M	
	→ 12 page faults — ⑤ M	

7 a)

*) Load the pages only as they are needed. ① M

*)



⑥ M

*) Above 6 steps explination. ③ M

7 b)

(i) Optimal — ⑤ M ← Efficient

7	0	1	2	0	3	0	4	2	3
7	7	7	2		2		2		
	0	0	0		0		4		
		1	1		3		3		

0	3	2	1	2	0	1	7	0	1
2			2				7		
0			0				0		
3			1				1		

→ 9 page faults

⑤ M

(ii) LRU — ⑤ M

7	0	1	2	0	3	0	4	2	3
7	7	7	2		2		4	4	4
	0	0	0		0		0	0	3
		1	1		3		3	2	2

0	3	2	1	2	0	1	7	0	1
0			1		1		1		
3			3		0		0		
2			2		2		7		

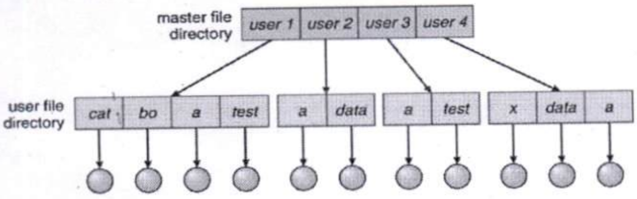
→ 12 page faults

⑤ M

Subject Title : Operating System   Subject Code : 21CS44

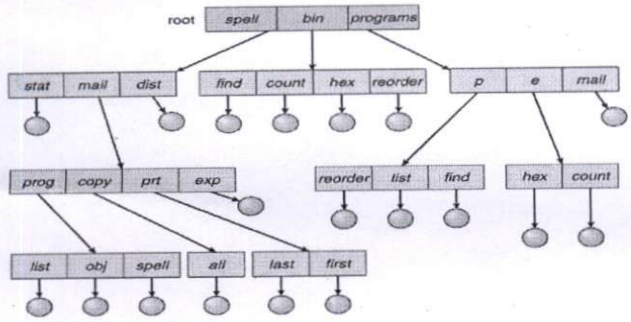
Question Number	Solution	Marks Allocated

8 a) *> Two-level directory structure — Marks: 8 (4+4)



with explination — ④ M

*> Three-level directory structure



with explination — ④ M

8 b) (i) Contiguous disk Space allocation — Marks: 12 (6+6)

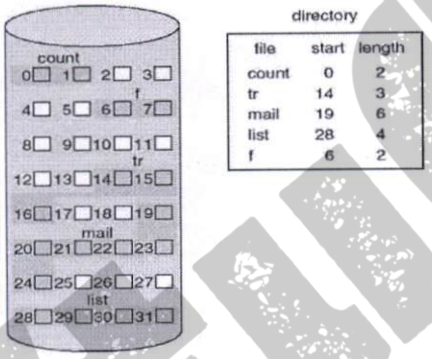


diagram → ④ M

Explination → ② M

(ii) Linked disk Space allocation

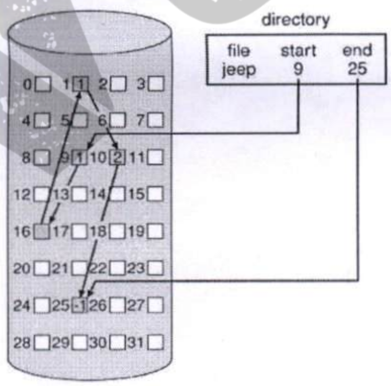
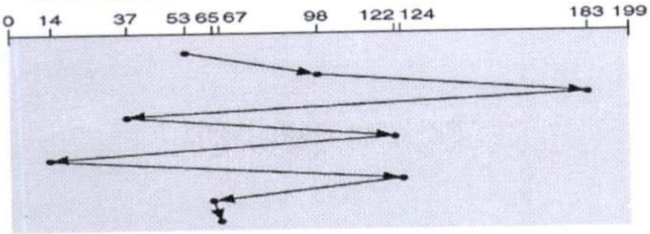


diagram → ④ M

Explination → ② M

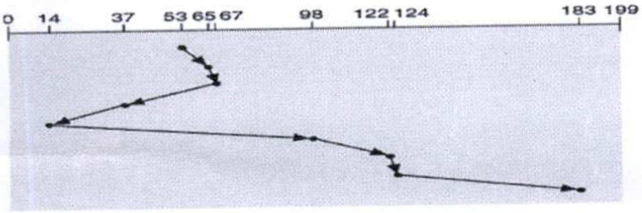
Subject Title : Operating System **Subject Code :** 21CS44

Question Number	Solution	Marks Allocated
9 a)	(*) FCFS	12 (3+3+3+3)



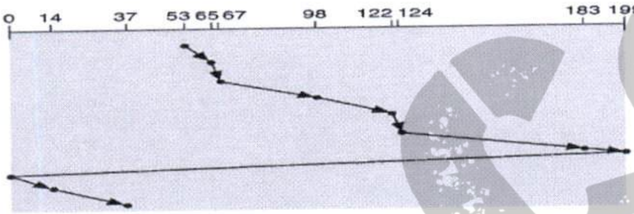
③ M

(*) SSTF



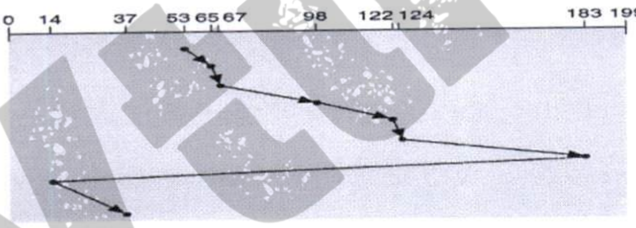
③ M

(*) C - SCAN



③ M

(*) C - LOOK



③ M

Question Number	Solution	Marks Allocated
9 b)	Access matrix model	8 (4+4)

Domain \ Object	F_1	F_2	F_3	Laser Printer
D_1	read		read	
D_2				Print
D_3		read	execute	
D_4	read/ write		read/ write	

Access Matirx

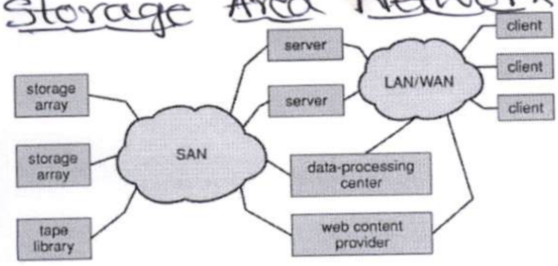
④ M

Explination → ④ M

Subject Title : Operating System Subject Code : 21CS44

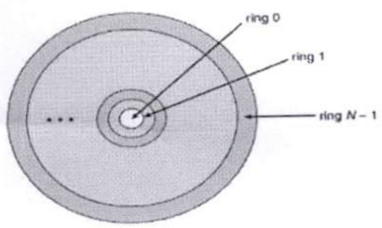
Question Number	Solution	Marks Allocated

10 a) (*) Storage Area Network (SAN) — 8 (4+4)



④ M

(*) MULTICS ring structure



④ M

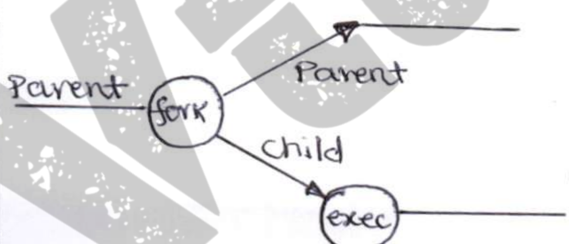
10 b) Components of a Linux System — 6

system management programs	user processes	user utility programs	compilers
system shared libraries			
Linux kernel			
loadable kernel models			

with explanation

⑥ M

10 c) fork and exec in Linux — 6 (4+2)



A process executes the fork system call, which creates a new child process. The child process then exec's the program to be executed.

④ M

ex: int pid

```
if ((pid = fork()) == 0) {
    execve("child", .. , ...);
}
```

② M